

Rocket Reusability and the Conditions for Long-Term Cost Reduction

Lakshya Kolli

Centennial High School

Intern/Mentor Program

Ms. Toni Ireland

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Abstract

Rocket reusability is widely thought of as a guaranteed method of lowering the cost of space flight. But, its benefits depend on the combination of a variety of factors, such as how the vehicles are designed, recovered, and operated. This paper examines how rocket reusability can help in lowering long term launch costs, and what determines if reusable launch vehicles (RLVs) are economically viable. Using technical reports, academic studies, and cost modeling methods, this study found that RLVs were most successful with high launch frequencies, low refurbishment costs, quick turn-around times, and reliable recovery systems. The study also found that payload size, vehicle design, certification requirements, and environmental impacts all influence whether it could be cost effective compared to expendable launch systems. Overall, rocket reusability can reduce long-term launch costs, but only when engineering design, operational efficiency, safety verification, and market demand are balanced together.

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Introduction

A single rocket launch can cost hundreds of millions of dollars, yet the vehicle is discarded after one use. Expendable launch systems incur high mission costs as they are discarded after each flight (Webb, 2016). These components are aimed to be recovered and reflown through reuse, but this brings in refurbishment and inspection demands, along with safety certification and reliability validation in between operations (De Oliveira & Lavagna, 2024; Li et al., 2025; Sagliano et al., 2025). Overall, the benefits gained through pursuing rocket reusability only occur once engineering design, operational efficiency, guidance reliability, and environmental considerations are carefully balanced (Dominguez Calabuig et al., 2024; Ryan et al., 2022).

Rocket reusability has the potential to significantly reduce long term launch costs, but these reductions are dependent on the interaction between design, operations, and verification requirements. This paper evaluates how reusability can reduce long-term launch costs, how design and operational decisions shape cost effectiveness, how frequent and safe reuse depends on recovery and reliability, and how environmental impacts vary with launch frequency and fuel choice.

Review of Literature

Cost Reduction and Economic Viability

Private companies/states, the mission, and the competitive nature of each are limited by their high launch costs. Reusability is centered on the goal of using the same part multiple times, so the cost of bringing the reusable hardware back into flight must be less than the traditional cost of replacing the hardware. Refurbishment, inspection, documentation, and readiness activities contribute to the overall cost of returning the hardware to flight. A significant amount of the cost savings from reusability comes from how often those reuse cycles happen so that the savings build over time.

The biggest factor limiting both the reuse aspect of reusable launch vehicles (RLV) and the economic impact of reuse is the cost and time associated with refurbishment and turnaround. In order to keep the average launch cost at an acceptable level, the refurbishment and turnaround time must be outweighed with extensive reflight. It is stated that under frequent operations, costs have been reduced by approximately 52-75% in medium and heavy payload ranges due to reuse (Kang, 2025). To maintain sustainability, the operational frequency needs to be high enough to distribute the fixed costs and stabilize the refurbishment workflow processes.

In addition to optimizing for individual launch costs, cost should also be considered within the context of the entire life cycle of the RLV. Costs associated with the development, production, refurbishment, and operation of the vehicle should also be accounted for. It is also important to understand that a simple design or low mass does not necessarily equate to cost-effectiveness. The outcomes from the Falcon 9 and the Space Shuttle demonstrate the importance of these factors. Although the Space Shuttle did not meet the target number of launches per year nor the projected costs, the Falcon 9 demonstrated the ability to achieve lower

costs per flight and increased launch frequency. This was a result of the direct consideration of the service life limits of the RLV, and the entire life cycle of the RLV (Dorney, 2022). This comparison also shows how reuse is prone to failing economically when factors such as inspection, refurbishment, and operational complexity prevent rapid, repeatable reflight.

As reuse goes on, certification and verification requirements become increasingly important. It is noted that requirements only in fracture control add significant cost through mandated analysis, testing, inspection, and documentation (Leser et al., 2025; National Aeronautics and Space Administration, 2024). This burden has the potential to offset cost savings if not refined. These findings show that a conditional cost model is important, with long term costs only being reduced when verification remains manageable.

Acquisition and program structure also impact cost outcomes greatly. Cost reductions in NASA's COTS/CRS program compared to the Space Shuttle resulted in cargo delivery costs that were estimated to be roughly 2-3 times lower per kilogram, and Falcon 9 development costs were estimated to be up to roughly 10 times lower than traditional cost-plus programs (Zapata, 2017). Additionally, NASA management costs were reduced from ~13% to ~5% of total program funds, showing the potential for these factors alone to make a large impact.

Design and Engineering Strategies for Reusability

Engineering decisions at the conceptual design level also have a key role in the success of the reusable launch vehicle. The key point about reusability is that it is not ideal for the rocket to

be optimized for payload or ascent performance because these design decisions made during the conceptual phase may cause the rocket to be too complex, too heavy, or too difficult to inspect/refurbish efficiently after recovery (Childress-Thompson, 2016; Dresia et al., 2021). Thus, the economic value of reusing a rocket is based upon both the number of times a rocket flies, and whether the original design provides a practical opportunity for repeated use (Childress-Thompson, 2016).

NASA's Reusable Launch Vehicle Technology Program (RLVTP), which provided funding for research and technology development on reusable rockets, emphasized the need for architectures that support reusability (e.g., Single-Stage-To-Orbit [SSTO] designs) but these architectures require advanced propulsion systems, lightweight structural components, reusable cryogenic tanks, and durable thermal protection systems (Cook, 1995).

Similarly, recent studies have shown that the technical feasibility and economic viability of a reusable rocket over time depend upon many factors including the type of propellants selected, structural mass, staging strategy, and recovery fuel margin (Dresia et al., 2021; Jo & Ahn, 2022). Therefore, cost-effectiveness of a reusable rocket is dependent upon the total life cycle performance, the recovery strategy, and the burden of refurbishment of the rocket rather than simply the mass of the rocket (Dresia et al., 2021).

A comparison between the Space Shuttle and the Falcon 9 shows how different architectures affect the reusable performance of a system. Although the Space Shuttle was designed with reusability in mind, the complexity of the design resulted in higher inspection,

maintenance, and turnaround costs compared to the more operationally manageable design of the Falcon 9. The Falcon 9 has been able to provide faster, more repeatable reflight opportunities due to its simpler design (Dorney, 2022). In turn, the design simplicity of a reusable system is likely to be a major factor in supporting faster and more repeatable reflight opportunities.

The cost to develop and manufacture landing systems, structural reinforcements, and recovery margins will increase compared to expendable systems. Payload capacity may be reduced alongside this, but these elements must be integrated within the conceptual design phase for the hardware to survive repeated flight cycles. (Jo & Ahn, 2022; Dresia et al., 2021). Therefore, reusability is an engineering optimization problem rather than a singular design feature.

Structural safety and certification are additional limitations on reusing spaceflight hardware. In order to be reused, a piece of space flight hardware has to perform well on cyclic thermal loading, mechanical loading, and repeated use, all while still fulfilling all of the qualification/verification requirements. The fracture control standards for NASA show how the classification of the hardware is related to the design margin, inspection strategy, and feasibility of repeated use (National Aeronautics and Space Administration, 2024). These requirements mean that structural decisions directly determine how many times a RLV can be reflown in its service life.

Lastly, the chosen design also affects the environmental performance of a reusable launch vehicle. These design decisions include vehicle architecture, propellant selection, and the method

used to recover the reusable portion of the launch system. Systems powered by hydrogen fuel may have significantly less negative impacts on climate, water, and land use compared to methane fueled systems. However, the emissions produced during launch and re-entry will still generate significant amounts of nitrogen oxide, and other atmospheric pollutants (Calabuig Dominguez et al., 2024; Ryan et al., 2022). Overall, the success of a reusable launch vehicle requires that each component is designed with consideration to the entire system's architecture, recovery strategy, structural design, manufacturing methods, certification, and environmental sustainability at the onset of development.

Operations, Refurbishment & Reliability

Whether a reusable launch system can achieve the true promise of cost reduction depends upon more than just being able to fly again after one flight. It also depends on efficient turnaround times after each mission. Other operational considerations include: the speed at which reusable hardware can be inspected and maintained, the degree to which reusable hardware can be accessed for maintenance, the burden of inspection per flight, and the cost of labor for maintenance activities (Childress-Thompson, 2016). If the post-flight inspection effort is overly burdensome or hardware cannot easily be accessed for repair, then the benefits of reusability can be rapidly diminished. Therefore, reusable systems should be designed with the goal of minimizing the amount of work required between missions.

NASA's previous efforts with reusable launch vehicles have similarly recognized that the key to lowering launch costs in operational practices was to mirror those of commercial airlines. Rather than labor intensive refurbishment, NASA has identified short turn-around times, small

ground crews, automated check-out, health monitoring and maintenance strategies based on reliability and supportability as key elements to achieving cost reduction (Cook, 1995). Unlike commercial aviation, which benefits from decades of codified maintenance procedures, reusable launch vehicles are still developing equivalent standards in real time. "Aircraft maintenance has 70 years of procedures, and reusable rockets are literally rewriting all of these processes in real time" (S. Sethi, personal communication, February 12, 2025). This brings in the hardware and operations problems that must be considered with reusability. A vehicle may be capable of being recovered, but a reuse model will be far less effective if the ground processing following recovery remains slow and costly.

The history of operations further supports the idea that maintenance strategy plays a critical role in determining whether a reusable system is successful. Dorney's analysis of the Space Shuttle and Falcon 9 has shown that reusable systems benefit from structured maintenance plans, life tracking, defined inspection intervals, and well-defined certification processes throughout the life of the vehicle. Specifically, Dorney demonstrated that Shuttle operations were characterized by long turnaround times and significant labor requirements, while Falcon 9 benefited from more precise inspection criteria and staged acceptance processes. Dorney's comparison demonstrates that simple operation and discipline in maintenance practices play a critical role in whether reusability can ultimately lead to cost savings.

Another major issue is that refurbishment itself can become one of the largest contributors to total life cycle cost. If much of the money was used on repairing a rocket at the end of each flight, it could be difficult to make up for the expense of the additional features

needed to recover a rocket and to design rockets to fly again. By reducing the stresses on the rocket's structure during both ascent and descent, the amount of work that needs to be done after each flight can be reduced significantly. This shows us that operation and design have to stay in close contact with one another, and reuse is only cost-effective if the hardware being reused doesn't require extensive repairs, tests, or inspections between flights (Childress-Thompson, 2016; Dorney, 2022).

In addition to the design of reusable systems, certification requirements also add significant operating burdens. NASA has a fracture control standard that all NASA programs must comply with. For hardware built for repeated use, the fracture control standard requires that programs establish fracture control plans. These can include items such as inspection cycles, life tracking, verification testing, and review oversight (National Aeronautics and Space Administration, 2024). Similarly, NASA has used additive manufacturing to produce parts for spaceflight hardware, showing that a number of steps are needed to ensure safe reuse, including statistical process control, witness testing, inspection and repeated life tracking (Russell et al., 2022). While these requirements are important for ensuring reliability, they also add to the work that needs to be done before the next flight, and can reduce the expected cost savings from reusable operations.

Reusable operations also have broader implications for sustainability and program management. As explained by Dominguez Calabuig et al., although the majority of the lifecycle footprint of reusable launch vehicles includes recurring activities such as launch emissions, factors such as recovery operations, refurbishment, and maintenance can limit the amount of

environmental benefit that can be achieved through reuse alone. The COTS/CRS model analyzed by Zapata demonstrates that fixed price partnership structures can result in more efficient and reliable long-term operations by transferring some of the financial risks associated with those operations to the partner(s), and encourage repeatable commercial performance (Zapata, 2017; Dominguez Calabuig et al., 2024). Altogether, this shows that the value of reusability depends not just on reflight itself, but on whether the vehicle can be maintained, certified, and operated repeatedly without creating excessive cost, delay, or operational burden.

Guidance, Recovery, and Flight Performance

Reusable launch systems can only provide long-term value if the hardware they return from spaceflight is in good condition to use again. Thus, recovering a rocket stage with minimal or no significant damage to hardware is one of the key requirements to achieve reusability. The systems required to recover hardware include landing hardware, recovery reserve fuels, and control systems. These systems also add to the overall vehicle weight and decrease the rocket's payload. Therefore, reusability will always require a trade-off in terms of rocket performance since some rocket capability must be allocated to bring hardware back rather than maximize payload delivery (Childress-Thompson, 2016).

Previous NASA reusable launch vehicle programs clearly showed that autonomous flight control, re-entry capabilities, and landing performance were critical technical objectives for reusable systems. Demonstrations of these flight functions included vehicles such as DC-XA, X-34, and X-33. It was determined through these demonstrations that reusable launch vehicles would have to demonstrate flight functions reliably under actual mission conditions and not just

theoretical conditions (Cook, 1995). This displays that recovery is not simply an add-on to a rocket, but a significant flight-performance challenge that has always been a primary objective of reusable vehicle development.

Recent research continues to reinforce this concept by demonstrating that precise landings depend upon the integration of advanced guidance and control. For example, De Oliveira and Lavagna developed a descent system that uses successive convex optimization to generate guidance trajectories and use robust control to ensure trajectory tracking, while disturbances such as winds, aerodynamic uncertainties, and actuator limitations are present. Results indicate that their method provides improved landing accuracy and increased recovery reliability. Successful recovery is enhanced when the guidance remains stable in the presence of realistic non-ideal conditions. Reliable reuse requires precise flight performance during the most difficult phases of flight; i.e., descent and landing (De Oliveira & Lavagna, 2024).

Recovery strategy impacts the total reusable system's performance. As different recovery methods (i.e., down-range landing, return to launch site, etc.) have varying effects on vehicle performance, re-use rates, and environmental concerns (Dominguez Calabuig et al., 2024), there are unique performance trade-offs associated with recovery requirements which need to be considered at the time a vehicle is being developed (e.g., propulsion during landing and aerodynamic control during descent). Jo and Ahn's life-cycle cost analysis supports this concept by indicating that reuse and recovery assumptions must be integrated into staging optimization from the beginning of design, as mass penalties and performance losses associated with recovery must be weighed against reduced manufacturing costs as a function of multiple flight cycles (Jo

& Ahn, 2022). This relationship demonstrates that flight performance and long-term cost-effectiveness are tied to both guidance and recovery.

In addition, although recovery capabilities and mission acceptability are correlated, they are not equivalent. Synthesis materials indicate that NASA's launch service evaluation procedures separate entry, descent, and landing performance from re-use, and place significant emphasis on qualified service life and risks associated with ascent. Thus, while demonstrating the ability to recover a stage is necessary for reuse, it does not ensure mission acceptance. Hence, the re-usability of a vehicle is dependent on ensuring that the hardware operates within acceptable risk and service life limits along with the landing the hardware (Dorney, 2022).

Thus, flight performance, guidance, and recovery are what allow the theoretical concepts of re-use to become a viable operational system. Although a re-usable rocket may reduce the amount of material required to manufacture new rockets, it will not be able to realize the benefits of lower manufacturing requirements until it has demonstrated the capability to consistently descend, land, and be recovered. The mass and payload penalties associated with recovery demonstrate that re-use is not cost-free. However, advancements in flight guidance and recovery design provide the basis for absorbing these penalties. Because of this, safe recovery and precise landing are not secondary features of reusable launch vehicles; they are some of the main conditions that determine whether repeated reuse is possible at all (Childress-Thompson, 2016; Cook, 1995; De Oliveira & Lavagna, 2024; Jo & Ahn, 2022).

Research Methods and Data Collection

To address the hypothesis, this paper utilized a comparative document analysis of academic and technical literature related to the economics of reusable rockets. The main purpose of the primary research was to determine if rocket reusability can lower long term launch costs, and what specific conditions have to exist before cost savings can occur. The analysis consisted primarily of comparing two criteria across the various sources. These include each source's requirements for achieving a launch cadence that would make reuse economically viable, and any variables that may shape cost such as payload size, reuse assumptions, recovery method, and refurbishment burden. Data were collected from each source by reviewing their stated objectives, methodologies, assumptions, tables, figures, and conclusions to allow for comparisons across the multiple studies.

Results and Data Analysis

Data was collected to evaluate whether rocket reusability reduces long-term launch costs under specific operational conditions. This data places emphasis on how the cost savings of rocket reusability depend less on the idea of reusing rockets, and more on the specific conditions under which reuse will be accomplished. In the five studies reviewed, the primary factor determining the economic feasibility of launch vehicles was found to be launch rates. Wertz noted that launch rate is one of the most significant long term factors affecting the total costs of launching a new spacecraft or satellite. It was further stated that the number of viable markets for a new launch vehicle would likely be in the range of 10 to 15 launches per year, a number at which expendable launch vehicles have an economically advantageous position over reusable launch vehicles. A similar conclusion was reached by Gstatenbauer who determined that expendable launch vehicles are preferable at lower launch frequencies and only begin to lose

their advantages over reusable launch vehicles once launch demand increases a large amount over time. It was also demonstrated by Kang that reusable launch costs decrease significantly as the annual launch frequency (especially in larger payload classes) increases.

Table 1.

Launch-Cadence Findings Across Reusable Launch Vehicle Cost Studies

Source (with author, publication date, etc.	Study design: methods, assumptions, timeframe, results	Annual launch cadence threshold for economic viability
Wertz (2000), Economic Model of Reusable vs. Expendable Launch Vehicles	Compared reusable and expendable LEO launch vehicles for a 5,000 kg payload over 2001-2015, using 15-year amortization, 15% interest, and an 87.5% learning curve.	At the expected market level of 10-15 launches/year, expendable vehicles remain cheaper. New expendables cost about \$10-35 million/launch, while reusable vehicles cost about \$35-90 million/launch.
Gstattenbauer (2006), Cost Comparison of Expendable, Hybrid, and Reusable Launch Vehicles	Compared ELV, HLV, and RLV systems over a 20-year system life, assuming a maximum reusable vehicle life of 200 launches.	In the full-development case, ELVs are preferred below 75 total launches, HLVs from 75-200, and RLVs above 200 over 20 years. In the real-world case, the hybrid system is preferred around 200-260 launches, while RLVs become more economical above about 225 launches over 20 years.
Kang et al. (2025), Cost Effectiveness of Reusable Launch Vehicles Depending on the Payload Capacity	ROM/CER-based model dividing total launch cost into development, production, reuse, operations, fixed costs, and insurance, with assumptions about reuse count, turnaround, and payload penalty.	For a 20-ton payload, reusable cost is about 64% of expendable cost at 20 launches/year and 52% at 200 launches/year. Break-even occurs at about 18 launches/year for 2 tons and 88 launches/year for 500 kg.
FALCon D2.1 (2020), Technical Report on Different RLV Return Modes' Performances	Compared reusable return modes using TRANSCOST and an in-house recovery model, with RLV costs normalized against comparable ELVs.	Increasing launch rate from 15 to 30 launches/year reduces recovery cost by about 30% for the SpaceX-style method, 40% for the Blue Origin-style method, and 35% for IAC.
Koelle and Janovsky (2007), Development and Transportation Costs of Space Launch Systems	Compared ELV and RLV transportation costs for GTO missions using 30 years of operation, a 0.875 learning curve, and different payload capacities.	At 1 flight/year, cost per flight rises by about 3× for the ELV and 9× for the RLV compared with nominal values. At higher launch rates, the RLV gains an estimated €70 million cost advantage, although the ELV carries about 2× the payload.

The results reinforce that the frequency of launch does not solely determine the cost effectiveness of using reusable versus expendable launch vehicles (Table 1). Payload size, assumptions regarding reusability, and the extent of refurbishment required also influence the degree that reusable systems provide cost reductions (Table 1). For example, Kang's study demonstrates that reusable launch cost reduces to approximately 64% of the cost of expendable rockets at twenty launches per year, and approximately 52% at two hundred launches per year for a 20 ton payload (Table 1). However, for a 2 ton payload, reusability only provides clear cost savings when the annual launch frequency exceeds approximately seventy launches per year (Table 1). Similar to Kang, Gstatenbauer concluded that larger payloads cause hybrid and reusable systems to be preferred at earlier launch frequencies. Finally, FALCon extends this trend by demonstrating that the economic viability of reusable systems also depends upon the conditions of recovery and refurbishment because reusable launch costs remain highly sensitive to these operational expenses. Overall, the results collectively support the paper's thesis with some limitations: the reduction in long term costs from reusability only occurs when combined with sufficient flight demand and operational conditions. These conditions must not allow recovery and reuse to become excessively expensive.

Table 2.*Cost-Shaping Conditions Affecting Reuse Economics*

Source (with author, publication date, etc.)	Cost-shaping conditions that affect whether reuse is advantageous	Ultimate Assessment
Wertz (2000), Economic Model of Reusable vs. Expendable Launch Vehicles	Results depend on whether the reusable vehicle reaches 100-200 flights per vehicle, along with recovery cost, refurbishment cost, insurance, 15-year amortization, and 15% interest.	Shows that cadence is central, but reuse is not automatically cheaper. At 10-15 launches/year, reusable vehicles still remain more expensive than expendables under the study's assumptions.
Gstattenbauer (2006), Cost Comparison of Expendable, Hybrid, and Reusable Launch Vehicles	Payload size, system life, vehicle life, and total launches determine the threshold. The study models a 20-year life and maximum reusable vehicle life of 200 launches.	Shows that reuse requires sustained long-term demand. RLVs become preferable only around 200+ total launches, meaning roughly 10+ launches/year over a 20-year period.
Kang et al. (2025), Cost Effectiveness of Reusable Launch Vehicles Depending on the Payload Capacity	Payload capacity is the strongest modifying factor. The model assumes 10 reuses, 30-day turnaround, 30% payload penalty, and 3% refurbishment/production cost.	Gives the strongest numerical support: for 20 tons, reuse is beneficial by 20 launches/year, but for 500 kg, break-even requires about 88 launches/year.
FALCon D2.1 (2020), Technical Report on Different RLV Return Modes' Performances	Refurbishment factor, recovery method, reuse count, and launch frequency shape cost. The RLV becomes cheaper when refurbishment factor drops below 0.4; at 0.25, it becomes cheaper at sufficiently high reuse counts.	Shows that high launch cadence is not enough by itself. Even with reuse, refurbishment and recovery costs can determine whether savings actually occur.
Koelle and Janovsky (2007), Development and Transportation Costs of Space Launch Systems	Payload size and annual launch mass are central. ELVs are more cost-effective below about 10 tons, costs are roughly equal around 9 tons/year to GTO, and RLV advantage grows at higher demand.	Supports the conclusion that reuse works best at larger scale. At higher demand, the RLV advantage is about 6-8 k€/kg, or roughly 20-50%, depending on payload and launch mass.

These results validate that the economics of reusing rockets are dependent upon the system being used (Table 2). This supports that launch vehicle development will need to take

additional account if there is sufficient long term market demand to justify the additional development cost, and if the vehicle can re-flown quickly between each flight (Table 2). In other words, this research finds that reusable launch systems would likely have the greatest potential in markets requiring high frequency of launch, and/or having greater payload requirements, whereas smaller launch systems would appear to have little or no economic justification to use reusable components (Table 2).

Additionally, this research contains some limitations. The first limitation is that the number of sources analyzed was limited. Since this analysis relied solely on five model-based analyses, the reliability of the findings were largely dependent on the accuracy of the modeling assumptions made by each researcher. For example, Kang states his estimates are ROM (rough order of magnitude), whereas FALCon indicates that accurately estimating reusable launch costs is complicated due to both the lack of reliable cost data available and significant uncertainty associated with many variables involved. Finally, none of the researchers studied exactly the same system. Wertz modeled a 5,000kg LEO capable rocket, Gstattenbauer developed models using hydrocarbon fueled TSTO rockets operating for twenty years, Kang compared various payload class sizes, and FALCon focused on two return methods and normalized launch costs. These differences limit the ability to compare them directly, but they also indicate possible areas for future research.

Discussion and Conclusion

The study's results demonstrate how the findings of this research add to the existing body of literature examining reusable launch economics by showing that reusing launch vehicles is not necessarily cheaper than using expendable launch vehicles. The studies reviewed show that a reusable launch vehicle only becomes more cost-effective under certain operational conditions, including high launch frequency, sufficient payload demand, low refurbishment costs, and efficient recovery operations. These conditions show that the economics of reusability depend on more than reuse itself. The repeated use must offset the higher development, recovery, and maintenance costs.

These findings have important implications for commercial launch providers and government space organizations. Commercial launch providers need to determine whether launch demand is sufficient to justify development and operating costs. Government agencies or private sector organizations comparing reusable and expendable systems should consider payload size, launch frequency, refurbishment needs, and recovery methods before assuming reusable systems will reduce costs. The literature indicates that while reusability is beneficial, it is most useful as an element of an overall economic system rather than as a stand-alone technology.

The most significant drawback of this methodology is its reliance upon model-based analyses and limited real-world launch cost data. Several authors reference this limitation. Kang refers to his estimate as a "rough order of magnitude," and FALCON notes that reusable launch cost estimations remain problematic due to the scarcity and uncertainty of reliable reusable cost data. Gstatenbauer states that maintenance cost estimates for hybrid and reusable vehicles are

uncertain since there is no large database of comparable operating systems. Additionally, none of the references examined identical systems. Wertz modeled a 5,000 kg LEO vehicle; Gstatenbauer compared hydrocarbon-fueled TSTO systems over a twenty-year life cycle; Kang studied multiple payload classes in LEO; and FALCON focused on different reusable return modes and normalized launch costs. This difference limits one-to-one comparison among them, but strengthens the overall analysis by demonstrating that the same economic trend exists across their respective methodologies.

To better understand the relationship between reusability and cost savings, future research should use more operational data from reusable launch systems. As more data becomes available, researchers should examine refurbishment costs, turnaround times, vehicle lifetimes, recovery costs, and launch rates. This data would help researchers assess whether current cost-saving estimates align with actual operational data.

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